United International University
Department of Computer Science and Engineering

CSE 3421: Software Engineering Final: Fall 2025

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all five questions. The numbers on the right of the questions denote their marks.

1. Design **UML diagrams** using the **appropriate design pattern** for the following scenarios: [CO2]

   (a) FitRide is a fitness application that allows users to track their workouts. The app supports various workout calculation techniques, including calorie-based, time-based, and distance-based workouts. A user can switch between these workout calculation techniques at runtime depending on their fitness goal. The application should allow adding new workout calculation techniques in the future without modifying existing code. (4)

   (b) UniPay is a university-wide online payment system used for tuition fees, examination fees, and library fines. To avoid data inconsistency and transaction conflicts, the system must ensure that only one instance of the payment manager exists throughout the application lifecycle. All payment requests from different modules must be handled through this single instance. (4)

2. (a) MediTrack is a newly developed mobile health application. Across different screens of the app, the same actions are labeled and displayed differently—for example, the "Submit" button appears as "Save" on some pages and "Confirm" on others, and its color and position vary from screen to screen. When users search for an unavailable report, the app shows an empty screen without any message explaining the situation. If users enter incorrect login credentials, the app simply refreshes the page without showing an error message or guidance on how to correct the mistake. Additionally, while booking an appointment, users must remember previously entered details, such as the selected doctor and time slot, when moving between multiple screens, as the app does not display or retain this information. There is no Undo option as well. (5)

   Which of **Schneiderman's 8 Golden Rules** of Interface Design are **violated** in the above scenario? **Explain** each violation with proper **justification** and propose suitable **design improvements** to address them. [CO1]

   (b) **Describe** the **advantages and disadvantages** of the **two types** of user interfaces. [CO1] (3)

3. (a) A municipality plans to develop an online platform for city administration services. The platform will offer features such as citizen data management, property tax calculation and online payment, submission of service requests and complaints, employee management, and generation of administrative reports. The entire platform will be designed as a single, unified system, managed by one development team and deployed as a single unit. The number of users is moderate, and there is no need to independently modify or deploy individual features. (5)

   Which architectural **pattern** will be **best suited** for the software above? **Explain** with **proper reasons**.

   If, in the future, the platform needs to handle **10 times** more user load and requests, which architectural **pattern** should be **adopted** then and **why**? [CO2]

   (b) **Explain** the **components** of the **MVC** architectural pattern. [CO2] (3)

4. (a) A software engineering team is developing a smart traffic monitoring system for a city to reduce congestion and improve road safety. During testing, several engineers discover that the system sometimes gives incorrect traffic signals, which could cause accidents. Although this issue is documented, the team decides not to delay deployment because they believe the error will probably not happen often. One developer, who is aware that they lack sufficient experience with real-time systems, continues working on critical modules without asking for help from more experienced professionals. Another engineer privately alters performance test results to make the system appear more reliable, so the team's work will be viewed more favorably. When a junior colleague raises concerns about safety risks, the rest of the team dismisses the concerns and excludes the colleague from further discussions. (5)
Which of the ACM **principles** of ethics are **violated** in the above scenario? **Explain** with proper reasoning. [CO3]

(b) **Explain** the **difference** between **Client and Employer** and **Management** ethics principle. Give **an example** of **violation** for **each** of the principle. [CO3] (3)

5. (a) Find the **Earliest Start, Earliest Finish, Latest Start, Latest Finish, Float, Project Deadline, and Critical Path** using the **Critical Path Method** for the following tasks: [CO2] (8)

Tasks	**Duration**	**Precedents**
A	2	-
B	1	-
C	3	-
D	5	A
E	4	B,C
F	1	D
G	3	D,E
H	5	F,G
I	3	G
J	2	E
K	3	H,I